Hyperspectral Anomaly Detection for Spectral Anomaly Targets via Spatial and Spectral Constraints

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***Abstract*— Anomaly detection in a hyperspectral image (HSI) has been actively researched in the field of remote sensing due to its significant application requirements. Traditional methods were based on the spatial models for the background to detect the anomaly targets. However, in detecting the spectral anomaly targets, they led to two problems: 1) the spatial characteristics of spectral anomaly targets are not obvious, which causes many false alarms in detection and 2) spectral anomaly usually occurs in the local band of targets, while the rest of the spectrum is similar to the ones of surrounding backgrounds, which leads to missed detection. This article proposes a novel hyperspectral anomaly detection method for spectral anomaly targets based on spatial and spectral constraints (SASCs). This model finds suspected anomaly target part as spatial anomaly results through SASCs. Then, the feedback process determines the spectral anomaly through the spectral difference between the tested pixel and the surrounding background. It is fed back to the spatial anomaly results to obtain final detection results. Furthermore, in order to enlarge the spectral difference between anomaly and background while suppressing the background, the optimal order of fractional Fourier transform (FrFT) is determined by combining spatial anomaly results with the uncertainty principle, which is used in FrFT of HSI. Experimental results show that the proposed method suppresses the background and reduces the false alarm rate. The feedback mechanism effectively reduces the missing detection rate, achieving a promising detection accuracy.**

***Index Terms*— Anomaly detection, feedback mechanism, fractional Fourier transform (FrFT), hyperspectral images (HSIs), spatial–spectral constraints.**

I. Introduction

HYPERSPECTRAL images (HSIs) contain 3-D data, in which the spectral dimension captures abundant spectral information of the ground surface with the hundreds of narrow spectral bands, and the other two dimensions contain spatial information [1]. Compared with other remote sensing images, each pixel of the HSI has an approximately continuous spectral curve [2]. With such advantages, HSIs have been widely used in many fields, such as land object classification [3], military reconnaissance [4], mineral exploration [5], and target detection [6], [7]. The target detection aims to identify the interesting pixels based on either the prior spectral information of the target or the significant differences between targets and backgrounds. The target detection aiming to distinguish the difference between the target and backgrounds is also called anomaly detection [8]. It is considered more practical because no prior spectral information about the target is required. Accordingly, the HSI anomaly target detection can be widely used in practical applications: military target detection [9], forest-fire prevention [10], geological monitoring [11], and food health detection [12]. The traditional anomaly detection method determined the anomaly pixels based on the statistical characteristics between backgrounds and targets [13]. The most representative method is Reed–Xiaoli (RX) detector [14], which distinguished the anomaly pixels using Mahalanobis distance compared to the background pixels. A multivariate Gaussian distribution was assumed in modeling the probability density function of background pixels in the RX detector. A pixel with a larger Mahalanobis distance than a predefined threshold was classified as an anomaly. Otherwise, it was determined as a background. If the background distribution follows Gaussian distribution, the RX detector could provide a promising detection accuracy with low computational complexity [15].

The RX detectors are divided into two versions of implementations according to the background description: the local RX (LRX) and the global RX (GRX). The LRX takes the locally neighboring pixels as the background in a dual-window method for anomaly detection. In contrast, GRX regards the entire image as the background. All the pixels from the background region: either a local region or an entire image, are utilized to estimate the background statistics. Accordingly, the statistics: mean and covariance matrix can be easily contaminated by anomaly pixels, and thus the RX detector becomes often unstable. Furthermore, the actual HSI data are too complicated to model as satisfying the Gaussian distribution. In order to address those limitations, variants of the RX anomaly detector have been actively explored. A kernel RX (KRX) detection method expanded a linear low-dimensional non-Gaussian model to a higher dimensional nonlinear Gaussian feature space [16]. The cluster KRX (CKRX) [17] was proposed based on clustering, which reduced the high computational complexity of KRX while achieving comparable detection accuracy. The CKRX

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grouped background pixels into clusters and applied a fast eigen decomposition algorithm. In the subspace RX (SSRX), the RX detection was applied to the principal component analysis (PCA) partial band [18]. The background distribution of real images was well estimated in recent RX variants, such as iterative linear RX and the weighted RX (W-RX) [19], and segmented RX algorithms [20]. Also, several robust anomaly detectors were proposed, such as the blocked adaptive computationally efficient outlier nominator (BACON) [21], the random-selection-based anomaly detector [22], and the probabilistic anomaly detector (PAD) [23]. These methods utilized robust statistical strategies to prevent the contamination of anomalous signatures by removing anomalous signatures during the background estimation process.

The anomaly matrix is usually sparse, while the background matrix is low rank. Thus, an HSI can be decomposed into anomaly parts and background parts through the low-rank sparse reconstruction. This property can be utilized in anomaly detection [24]. For example, Ning *et al.* [25] proposed the potential anomaly and background dictionary construction (PAB-DC) method, which decomposed an HSI into the background (low-rank), anomaly (sparse), and noises for the hyperspectral anomaly detection. An HSI was divided into blocks, and each block is used to construct anomaly and background dictionaries under the joint collaborative representation (JCR) model. Then, the anomaly was detected using those dictionaries. Qu *et al.* [26], [27] proposed abundance- and dictionary-based low-rank decomposition methods for hyperspectral anomaly detection. In those methods, the low-rank constraint was applied on the background coefficients instead of the background itself, such that a significant amount of noisy pixels in the anomaly matrix can be removed. Wang *et al.* [28] formed a background dictionary with the material signature matrix for the LRR model to extract the background information for the anomaly target identification.

In addition to the statistical approaches described above, representation theory was also used in anomaly detection algorithms under the assumption that dictionaries can be used to represent HSI pixels. For instance, collaborative representation detection (CRD) [29] assumed that background pixels could be approximately represented by a linear combination of their neighboring pixels while anomaly pixels cannot, indicating that the background pixels and anomaly pixels can be distinguished. The background joint sparse representation detector (BJSRD) [30] measured the matched projection length of the tested pixel on the orthogonal complementary background subspace by using the joint sparse representation to detect anomalous pixels. The low-rank and sparse matrix decomposition (LRaSMD) method [31] modeled the anomalies and noise separately. However, it still could not effectively distinguish the weak anomalies from noise. Recently, Xie *et al.* [32] proposed an adaptive weighting detector based on structured tensor and guided filtering (STGF) for hyperspectral anomaly detection, which achieved better detection performances. The total variation and sparsity regularized decomposition model (TVSDM) [33] constructed the dictionary of background and potential anomalies by density peak clustering. In the low-rank sparse decomposition, the total variation regularization method was applied to reconstruct the background, and the sparsity-inducing regularizer was used to characterize the anomalies. Besides, Tao *et al.* [34] proposed an improved RX method, taking advantage of the fractional Fourier entropy (FrFE-RX). It could increase the noise suppression and discrimination capability of the detector.

The spectral anomalies, as well as the spatial anomalies, are included in the targets of HSI anomaly detection [35]. The spatial anomaly is mainly caused by the target contained in the HSI pixel, such as aircraft and tanks, resulting in the difference between the target pixel and the surrounding background pixels, which are well known and often appear in images. In contrast, spectral anomalies are mainly caused by "invisible targets" whose characteristics are not apparent in the spatial domain since it comes from spectral changes, especially occurring in the local spectrum band. They are mainly manifested as camouflage targets, hidden targets, and deceptive targets [13]. The typical spectral anomaly targets include artificial camouflage targets, submarines in ports, and burning points in forests. Due to their invisibility, detecting such spectral anomaly targets is challenging but essential. HSI can be used to detect spectral anomaly because it has abundant spectral information due to its high spectral resolution [36].

However, most of the existing anomaly detection methods aimed to address spatial anomaly detection, usually based on the background modeling scheme. Thus, a naive adaptation of the methods induces the following problems.

First of all, the existing anomaly detection algorithms often induce too many false alarms due to the ambiguity of the spatial anomaly of the spectral target [37]. Therefore, how to reduce the false alarms of spatial anomaly results through the analysis between the anomaly target and the background is a significant challenge. Second, the existing algorithms take less consideration of spectral anomaly targets, and the spectral anomaly occurs in the local range of the spectral curve, and the rest of the spectrum is consistent with the background, which can induce missed detection. Consequently, it is also challenging to enhance the spectral anomaly in anomaly detection of spectral anomaly targets.

Therefore, the existing methods often suffer from low accuracy and high false alarm rate. This article proposes a novel hyperspectral anomaly detection method for spectral anomaly targets based on spatial and spectral constraints (SASCs). In the proposed model, the spatial anomaly results with false alarms are obtained through spatial information constraints. The spectral constraint is used to determine the spectral anomaly results based on the spectral difference between the tested pixels and the surrounding backgrounds. The spectral anomaly results are fed back to and combined with the spatial anomaly results to obtain the final anomaly results. Furthermore, the spectral difference between anomaly and background is enlarged while suppressing the background. The spatial anomaly results are used as guidance information in which the optimal order of fractional Fourier transform (FrFT) is determined by combining spatial anomaly results with the generalized uncertainty principle, and the optimal order is used in FrFT of HSIs.

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The major contributions of this article are summarized as follows.

1) A novel spatial–spectral constraints method is proposed for hyperspectral anomaly detection. The spatial anomaly detection is carried out by statistical decision based on the similarity and difference between the target pixels and surrounding background pixels. The spectral constraint realizes spectral anomaly detection through the spectral difference between targets and surrounding backgrounds. Then, the spectral anomaly results are fed back to the spatial anomaly results to obtain final detection results. The experimental results prove that the proposed method can evidently improve the detection accuracy and provide an effective and robust detection performance compared with the existing methods.
2) In view of the fact that the spectral anomaly target is basically invisible in space, its difference from the background is reflected in the partial band range of the spectrum. This article proposes a new strategy to determine the optimal order by combining the spatial anomaly results with the generalized uncertain principle and applies it to the FrFT of HSIs as the intermediate process from spatial constraint to spectral constraint. The spectral anomaly targets are enhanced, and the backgrounds are suppressed, which is conducive to the detection process of spectral constraint.

The rest of this article is organized as follows. Section II describes the proposed method in detail. Section III discusses and analyzes the experimental results. Finally, the conclusion is drawn in Section IV.

II. Proposed Method

The proposed method consists of the following three steps. The first step is spatial information constrain through the spatial information of anomaly targets in HSIs. In the second step, the optimal order of the FrFT is determined by combining the spatial anomaly results with the generalized uncertain principle, which is used to suppress the background while highlighting the spectral anomaly. In the third step, the spectral anomaly results are obtained by spectral information constrain based on the spectral difference between the target pixels and the background pixels. Then, the spectral anomaly results are fed back to the spatial anomaly detection results while the final anomaly results are obtained. The overall algorithm is described in Algorithm 1, and the details of each step are explained in Sections II-A–II-C.

A. Spatial Constraint for Anomaly Detection

The spatial information is used to find the anomaly target candidates in the image. First, the spatial information between the observing pixels and the surrounding pixels is compared to determine the candidates [38]. The spatial relation between the anomaly target and its neighboring pixels is categorized into four cases, as shown in Fig. 1.

Case 1: The anomaly target pixel is surrounded by other anomaly target pixels, which occurs in the case of large target size.

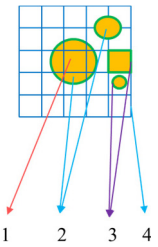


Fig. 1. Spatial relationship between the anomaly targets and the neighboring pixels.

Case 2: The anomaly target pixel is surrounded by some other anomaly target pixels and some background pixels.

Case 3: The anomaly target pixel is surrounded by the background pixels, which occurs when a single anomaly target occupies the whole pixel and subpixel target.

Case 4: There is no anomaly in the pixel, only the background, which is the case that there is no target in the pixel.

In the three cases that contain anomalies (Cases 1–3), the spectral vector of the pixel containing the anomaly target is correlated with the surrounding pixels. Therefore, the correlation between the anomaly target and the surrounding pixels can be used as a constraint to determine anomaly targets. The statistical characteristics of the surrounding pixels are used to determine the existence of anomalies. This statistical judgment process tests the null hypothesis H_0 that the target does not exist against the alternative hypothesis H_1 that the target exists

$$H_0\colon\ \mathbf{y} = \mathbf{v}\ \text{(Anomalous target absent)} \tag{1}$$

$$H_1\colon\ \mathbf{y} = \mathbf{D}\boldsymbol{\alpha} + \mathbf{v}\ \text{(Anomalous target present)} \tag{2}$$

where $\mathbf{v}$ and $\boldsymbol{\alpha}$ represent the background noise and the target spectral signal, respectively, and $\mathbf{D}$ is a signal coefficient.

The double window structure is used in spatial constraint. Let $\mathbf{X} = \{x^i\}_{i=1}^{M\times N}$ represent the 3-D HSI, where x^i is the ith spectral vector with P bands where each pixel is $x^i = [x_1^i, x_2^i, \ldots, x_P^i]$, and the size of $\mathbf{X}$ is (M, N, L), L is the number of bands. Let w_{in} and w_{out} be the size of the inner and outer windows, respectively. The number of atoms in the dictionary is s computed as follows:

$$s = \text{win}_{\text{out}} \times \text{win}_{\text{out}} - \text{win}_{\text{in}} \times \text{win}_{\text{in}}. \tag{3}$$

For a test pixel $\mathbf{y}$, the neighborhood pixel matrix $\mathbf{X}_s$ represents pixels between the outer and inner windows, where $\mathbf{X}_s = \{x^j\}_{j=1}^{s}$.

At this point, the problem is transformed into finding a weight vector σ, which makes the following constraint hold:

$$\min\|\mathbf{y} - \mathbf{X}_s\sigma\|_F^2 + \lambda\|\sigma\|_F^2 \tag{4}$$

where λ is the Lagrange operator. The corresponding Lagrange function is as follows [29]:

$$l = \arg\min\|\mathbf{y} - \mathbf{X}_s\sigma\|_F^2 + \lambda\|\sigma\|_F^2. \tag{5}$$

In order to increase the correlation constraint, the correlation matrix $\mathbf{C}_y$ is introduced [39]

$$\mathbf{C}_y = \begin{pmatrix} \|\mathbf{y} - x_1\|_2 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \|\mathbf{y} - x_s\|_2 \end{pmatrix}. \tag{6}$$

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Then, the Lagrange function is modified as follows:

$$l_2 = \arg\min \|\mathbf{y} - \mathbf{X}_s\sigma\|_F^2 + \lambda\left\|\mathbf{C}_y\sigma\right\|_F^2. \tag{7}$$

Taking the extremum of σ, the following equation is obtained:

$$\sigma = \left(\mathbf{X}_s^T\mathbf{X}_s + \lambda\mathbf{C}_y^T\mathbf{C}_y\right)^{-1}\mathbf{X}_s^T\mathbf{y}. \tag{8}$$

The results of background representation in the pixel $\mathbf{y}^*$ to be measured are as follows:

$$\mathbf{y}^* = \mathbf{X}_s\sigma. \tag{9}$$

Based on the binary hypothesis (1–2), the detection criterion based on the constant false-alarm rate (CFAR) of a measured pixel can be formulated as

$$\begin{aligned} r_1 &= (\mathbf{y} - \mathbf{y}^*)^T\mathbf{\Gamma}^{-1}(\mathbf{y} - \mathbf{y}^*) \\ &= (\mathbf{y} - \mathbf{X}_s\sigma)^T\mathbf{\Gamma}^{-1}(\mathbf{y} - \mathbf{X}_s\sigma) \end{aligned} \tag{10}$$

where Γ is the background covariance matrix. In this way, the anomaly detection results with spatial constraint r_1 are obtained and used to guide the FrFT of hyperspectral image X.

B. Fractional Fourier Transform Guided by Spatial Anomaly Results

The spectral differences between the measured pixels and the ground objects represented by the surrounding background pixels are considered the spectral constraint. The spectral difference usually appears in the local range of the anomaly target spectral bands, making it difficult to distinguish the target from the background. In order to enlarge the spectral difference, a new strategy to determine the optimal order is proposed based on the combining of the spatial anomaly results with the generalized uncertain principle. It is applied to the FrFT of HSI to facilitate the spectral constraint process.

When the hyperspectral image is expanded according to the spectral dimension, for each pixel x, the corresponding FrFT can be expressed as follows [34]:

$$x_p(u) = \frac{1}{L}\sum_{s=1}^{L} K_p(u, s)x(s) \tag{11}$$

where x_p is the result of the pixel x through p-order FrFT. The kernel K_p (u, s) is defined as

$$K_p(u, s) = \begin{cases} A_\alpha \exp[j\pi(s^2\cot\alpha - 2su\csc\alpha + u^2\cot\alpha)] \\ \alpha \neq n\pi \\ \delta(s-u), \alpha = 2n\pi \\ \delta(s+u), \alpha = (2n+1)\pi \end{cases} \tag{12}$$

where u and s are indices, n is the integer, and p is the fractional order of FrFT; α is the rotation angle, as depicted in Fig. 2. Note that it can be concluded that when $p = 0$, the x_p is the original spectral vector, and when $p = 1$, x_p is the result of the traditional Fourier transform, according to the following definition of α:

$$\alpha = \frac{p\pi}{2}. \tag{13}$$

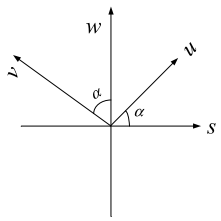


Fig. 2. Schematic of the relationship between FrFT and traditional Fourier transform.

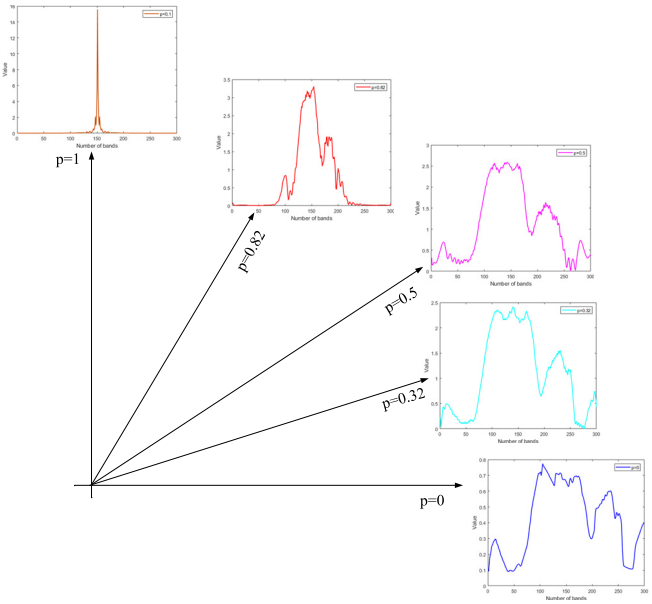


Fig. 3. Illustration of the influence of order p on the results of FrFT.

A_α is computed as follows:

$$A_\alpha = \frac{\exp\left[\frac{-j\pi sgn(\sin\alpha)}{4} + j\frac{\alpha}{2}\right]}{\sqrt{|\sin\alpha|}}. \tag{14}$$

When FrFT is applied to hyperspectral image anomaly detection, the optimal order p plays a critical role [40]. It can be seen from Fig. 3 that different order p has different effects on the pixels of the hyperspectral image. For hyperspectral anomaly detection with the spectral anomaly, the spectral anomaly can be highlighted while the background is suppressed with the optimal order p. In this article, the optimal order p is determined in two aspects: image information entropy and the spatial constraint results.

According to the uncertainty principle in the traditional time-frequency analysis field, a signal cannot simultaneously have an arbitrarily small resolution in the frequency domain and time domain, as defined as follows:

$$\Delta t \cdot \Delta w \geq \frac{1}{4}. \tag{15}$$

In essence, the principle of uncertainty not only explains the problem of hardware limitation but also gives a measure of the amount of information contained in the image, which corresponds to all the background and anomaly information contained in the image in anomaly detection.

Entropy is widely used as an image quality measure, reflecting the amount of information in the image. The Shannon

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entropy of an image $\boldsymbol{X}$ is defined as follows:

$$E = -\sum_{i=0}^{L-1} h_i \log(h_i) \tag{16}$$

where L represents the range of gray levels of $\boldsymbol{X}$, and h_i is the probability that the ith gray level occurs in the image. From the uncertainty principle, it can be seen that the time and frequency resolutions of the signal cannot be improved unlimited at the same time. Also, there is a lower limit for the product of the time and frequency domains. Time resolution and frequency resolution are a pair of Fourier transform, so they are the properties of the two transform domains. Formula (15) can be changed into the form of information entropy by the Shannon entropy formula (16)

$$E(|x_1(u)|^2) + E(|x(s)|^2) \geq \ln(\pi e) \tag{17}$$

where $x_1(u)$ is the conventional Fourier transformed signal of $x(s)$. Because the FrFT is obtained by rotating the coordinate axis on the basis of the traditional Fourier transform, the uncertain relationship between the time domain and the fractional domain needs to introduce the transformation order, that is, the rotation angle of the coordinate axis. Similarly, the FrFT of the rotating coordinate plane also has the corresponding uncertainty principle [40]–[43]

$$E(x_p(u)^2) + E(|x(s)|^2) \geq \ln(\pi e|\sin(p)|). \tag{18}$$

The FrFE can be obtained by applying the Shannon entropy operator to the FrFT [44]

$$\mathrm{FRFE}(p) = E \cdot \mathbf{X}_p \tag{19}$$

where $\mathbf{X}_p$ is the p order FrFT of $\mathbf{X}$, and $\mathrm{FRFE}(p)$ is the Shannon entropy corresponding to $\mathbf{X}_p$.

In FRFT, the order p is controlled to gather useful information by FrFT. The generalized uncertainty principle gives the lower limit of the optimization process. The lower limit represents all the information contained in HSI. For hyperspectral anomaly detection, the entire information contains the background and anomaly, and the anomaly information is useful. Since the information represented by the anomaly targets is not only the detailed information in the HSI but also the high-frequency information in the spatial domain, it is difficult to determine in a direct and accurate way. The spatial anomaly results are considered as suspected anomalies since they can include false detected ones besides true abnormal pixels. Therefore, the optimal order p is guided by the spatial anomaly results, which are used to determine the suspected anomaly information close to the anomaly information. The suspected anomaly information can contain a small amount of background information being false detected. The anomaly is finally determined by combining the spectral correlation in spectral constraint and the feedback mechanism process (Section II-C).

Fig. 4 depicts the relationship between the order p of the FrFT and the background/anomaly parts in HSI, where I represent background information. Parts II and III represent the suspected anomaly information from the spatial anomaly results; II is the false alarm in the spatial anomaly results and III

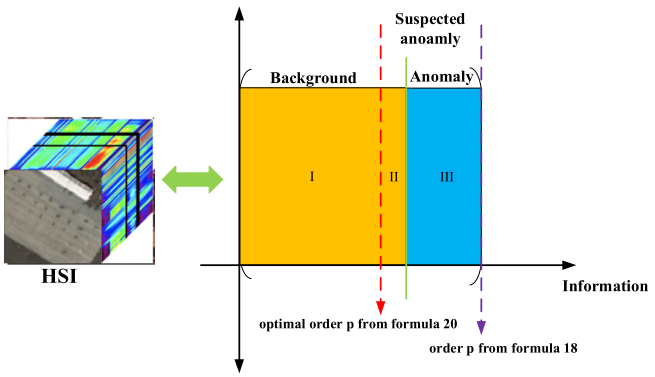


Fig. 4. Graphic method for determining the optimal order p.

is the anomaly information in HSI. From the perspective of anomaly detection, the information represented by the anomaly target is the most important. The sparse distribution of anomaly target pixels in the image shows strong disorder, so the corresponding amount of information is large. Taking the generalized uncertainty relation as a constraint [(18)], the obtained order p contains all the background and anomaly target information under the maximum fractional Fourier information entropy (the rightmost purple arrow in Fig. 4). The optimal order p can suppress most of the background information (the red arrow in Fig. 4), which is beneficial to the detection process constrained by spectral information.

The constraint conditions for determining the optimal order are obtained based on the generalized uncertainty principle and the spatial anomaly results. First, the representative pixel T of anomaly targets is selected by using the result of spatial information constraints. The specific method is to perform k-means clustering on the spectrum of anomaly pixels detected by spatial constraints. In this article, these pixel spectra are divided into k clusters. The selection of parameter k depends on different hyperspectral images, which is discussed in Section III-C. The clustering centroid containing the largest number of spatial anomaly pixels is selected as the representative pixel T of spatial anomaly results.

Accordingly, the generalized uncertainty principle and spatial anomaly representative pixel spectral T are applied to hyperspectral anomaly detection as follows:

$$E(x_p(u)^2) + E(|x(s)|^2) \geq \ln(\pi e|\sin(p)| - \eta \mathbf{T}^T \mathbf{T}) \tag{20}$$

where η is the adjustment factor, which is a small number. η is empirically set to 0.01. Then, the optimization function with the best order p is reformulated as follows:

$$\begin{aligned} l_3 &= \arg\max(\mathrm{FRFE}(p)) \\ &\text{s.t. } E(x_p(u)^2) + E(|x(s)|^2) \geq \ln(\pi e|\sin(p)| - \eta \mathbf{T}^T \mathbf{T}). \end{aligned} \tag{21}$$

Then, the hyperspectral X_p, which can fully concentrate the anomaly target information, is obtained by FrFT with the optimal p-value.

As described in Fig. 4, the FrFT guided by the spatial constraint results can extract the suspected anomaly part from the perspective of information, suppressing the background. The suspected anomaly part may contain a small background

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region, which is also required by the spectral constraint. The spectral anomaly can be detected using the relationship between the ground object spectrum.

C. Spectral Constraint and Feedback Mechanism Process

The optimal order of FrFT partially amplifies the local difference between the target and background spectrums. Then, in the spectral constraint process, the spectral difference between the pixel to be measured and the surrounding background pixel is compared to obtain the detection results. Finally, the spatial anomaly results are fed back to and combined with spatial anomaly results to obtain the final anomaly detection results.

The spectral relationship between the pixel to be measured and the ground objects represented by the surrounding background pixels is considered the main spectral constraint under the following assumptions [45]. If the candidate pixel is a background, it can be represented by the surrounding pixel spectrum, as described in the following equations:

$$M_0 : \mathbf{y} = \mathbf{A}\beta + \mathbf{n} \text{ (Anomalous target absent)} \quad (22)$$

$$M_1 : \mathbf{y} \neq \mathbf{A}\beta + \mathbf{n} \text{ (Anomalous target present)} \quad (23)$$

where $\mathbf{A}$ is the background feature dictionary represented by the pixel to be measured, and $\boldsymbol{\beta}$ is the coefficient corresponding to the background feature dictionary represented by the surrounding pixel, which is also called the corresponding dictionary atom. $\mathbf{n}$ represents the noise that is independent of the background and anomaly.

In HSI, for the pixel $\mathbf{y}$ to be measured, the corresponding coefficient $\boldsymbol{\beta}$ in the correlation hypothesis of the ground objects satisfy the non-negative and unique properties

$$\boldsymbol{e}^T\beta = 1 \quad (24)$$

$$\beta_i \geq 0, \quad i = 1, \ldots, N. \quad (25)$$

The spectral constraint is obtained by sparsely representing the part of the pixel $\mathbf{y}$ of the fractional Fourier transformed HSI and the spectrum of the background objects around it. The objective function is as follows:

$$r_2 = \|\mathbf{y} - \mathbf{A}\beta\|_2^2 \text{ s.t. } \boldsymbol{e}^T\beta = 1\beta_i \geq 0, \quad i = 1, \ldots, N. \quad (26)$$

The objective function is expanded as

$$\begin{aligned} r_2 &= \|\mathbf{y} - \mathbf{A}\beta\|_2^2 \\ &= (\mathbf{y} - \mathbf{A}\beta)^T(\mathbf{y} - \mathbf{A}\beta) \\ &= \beta^T\mathbf{A}^T\mathbf{A}\beta - 2\mathbf{y}^T\mathbf{A}\beta + \mathbf{y}^T\mathbf{y} \end{aligned} \quad (27)$$

where $\mathbf{y}^T\mathbf{y}$ can be ignored since it is a constant part of the objective function. Then, the objective function of object correlation is simplified as follows:

$$\min_{\beta} \boldsymbol{\beta}^T\mathbf{A}^T\mathbf{A}\beta - 2\mathbf{y}^T\mathbf{A}\beta \text{ s.t. } \boldsymbol{e}^T\beta = 1\beta_i \geq 0 \quad i = 1, \ldots, N. \quad (28)$$

The dictionary $\mathbf{A}$ in the objective function contains a small number of anomaly target candidates, making the residual result small to cause false detection and false alarm. This problem is caused by the fact that the pixels containing spectral anomaly are usually mixed pixels since a spectral anomaly is the change of the target spectrum. By controlling the coefficient value corresponding to the dictionary, the spectral anomaly part in the dictionary is separated from the background feature part, and the updated dictionary $\mathbf{A}^*$ and the corresponding coefficient $\boldsymbol{\beta}^*$ are obtained. The dictionary $\mathbf{A}^*$ and $\boldsymbol{\beta}^*$ represent the background features. Also, the mixed spectrum of the pixel may contain background features and anomaly target parts, and the regions with a higher abundance value are more likely to be anomaly targets. Thus, the parameter C is used to effectively distinguish the anomaly target in dictionary $\mathbf{A}$ and the background features, as shown in Fig. 5.

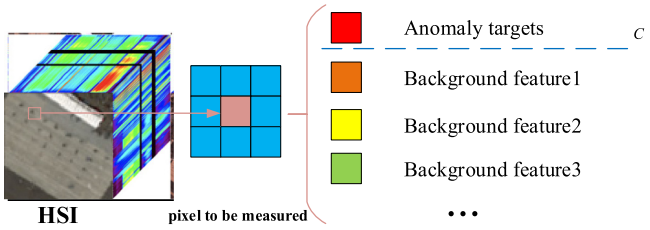


Fig. 5. Distinguishing background features from anomaly targets in spectral constraint.

When solving (28), the above problems are avoided through the increased residual error by removing the anomaly target candidates in dictionary $\mathbf{A}$. The coefficient $\boldsymbol{\beta}$ of the background part is

$$0 \leq \beta_i \leq C. \quad (29)$$

The range of C depends on the range of pixel value of HSI. The range of each pixel value is (0, 1). Accordingly, the range of C is (0, 1), and then the objective function can be transformed into

$$\min_{\beta} \beta^T\mathbf{A}^T\mathbf{A}\beta - 2\mathbf{y}^T\mathbf{A}\beta \text{ s.t. } \boldsymbol{e}^T\beta = 10 \leq \beta \leq C \; i = 1, \ldots, N. \quad (30)$$

This is a nonlinear (quadratic) optimization problem. Since the background dictionary, anomaly dictionary, and their corresponding coefficients have been obtained, the value of C can be determined by gradually narrowing the range.

Similar to the linear mixture model of HSI [46], the pixel to be measured is expressed as follows:

$$\mathbf{y} = \alpha_a\boldsymbol{a}_a + \sum_{i=1}^{p-1} \alpha_b^i\boldsymbol{a}_b^i \quad (31)$$

where $\boldsymbol{a}_a$ is the spectrum of the anomaly target, and α_a is the corresponding coefficient to the anomaly target. The $\boldsymbol{a}_b^i$ and α_b^i are the ith background feature and the corresponding coefficient, respectively. When α_a is equal to 1, the pixel $\mathbf{y}$ belongs to the anomaly target, while the coefficient range of α_a between (0, 1) indicates that the pixel $\boldsymbol{y}$ is a mixture of anomaly target and background features.

Usually, the solution of parameter C is very complex and challenging to determine. The approximate value of parameter C can be obtained by using the proportion of anomaly target in the hyperspectral image given as follows:

$$C \approx \frac{1}{\upsilon N} \quad (32)$$

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where N is the sum of the effective atomic numbers in the background and anomaly target dictionaries. Equation (32) can be intuitively explained that the pixel value in the HSI is normalized, and thus υ represents the proportion of anomaly target part in the pixel to be measured in the HSI.

For determining υ, first, the HSI data is normalized, as follows [47]:

$$\mathbf{X}_{\text{normalize}} = \frac{\mathbf{X}_{abd} - \min(\mathbf{X})}{\max(\mathbf{X}) - \min(\mathbf{X})} \tag{33}$$

where $\mathbf{X}_{abd}$ is the pixel value of ath row, bth column, and dth band in the HSI data. $X_{\text{normalize}}$ represents the normalized pixel value. The $\max(\mathbf{X})$ and $\min(\mathbf{X})$ denote the maximum and minimum values of the HSI pixel, respectively. Generally, the content of anomaly targets in the image is less, and the overall distribution is sparse, accounting for about 5%–25% of the image. The parameter υ of each image is mostly related to the content of the anomaly targets in the image. The optimal value of parameter υ can be determined through empirical experiments. The specific determination process is described in Section III-C.

After getting the value of υ, the corresponding C value is also determined. Then, the updated dictionary $\mathbf{A}^*$ and the corresponding atoms $\boldsymbol{\beta}^*$ are obtained through C. The influence of the anomaly target candidates in the background feature dictionary on the detection is significantly removed from these processes.

The detected anomaly by using spectral constraint is computed as follows:

$$r_2 = \left\| \mathbf{y} - \mathbf{A}^* \boldsymbol{\beta}^* \right\|_2. \tag{34}$$

Both SASCs between the pixel and the neighboring pixels are measured for the HSI anomaly detection. In order to fully utilize the spatial–spectral information for the anomaly target candidates, the two information constraints are combined

$$r = r_1 \leftarrow r_2 \tag{35}$$

where the $\leftarrow$ represents the feedback of r_2–r_1 in the same anomaly target pixel.

The main process of the feedback mechanism is as follows. First, the spatial anomaly pixels are regarded as suspected anomaly pixels. For each suspected anomaly pixel, the spectral anomaly detection results at the same position are used to determine it. If the spectral anomaly constraint is also detected as an anomaly, then the pixel is judged as an anomaly target. In contrast, if the spectral constraint is detected as a background, it is considered background. Finally, the final detection results are obtained.

Algorithm 1 The Proposed SASC Method for Hyper-Spectral Anomaly Detection

Inputs: 3D-HSI, a double window structure (win_{out} win_{in}), parameter $\eta = 0.01$, υ.
Initialize: Linearly normalize the whole image by (33).
Steps:
for all pixels, do
1. For each tested pixel $\mathbf{y}$, the spatial correlation is expressed in the double window structure by (9). Then, the detection result of spatial constraint is obtained through the statistical decision by (10).
2. The k-means clustering is used to select the representative pixels representing the anomaly target in the spatial anomaly results. The optimal order p is determined by the generalized uncertainty principle by (20). Then, the fractional Fourier transform is performed on the hyperspectral. The spectral information of the HSI after fractional Fourier transform is constrained. The spectral anomaly pixels are found using the difference between the measuring pixel and the surrounding background spectrum, according to (34). The spectral anomaly results are obtained and fed back to the spatial anomaly results to obtain the final detection results.

End for
Output: Anomaly detection map

III. Experimental Results

A. Hyperspectral Dataset

Six real hyperspectral datasets captured over different scenes are used to evaluate the detection accuracy of the proposed method. The details of the datasets are described in the following.

1) The San Diego Dataset is the first HSI dataset (HSI-I). The HSI data were collected in the San Diego airport area, CA, USA, by the AVIRIS sensor. Each band of the HSI data had 100 × 100 pixels, and the noise bands were removed. There are 38 helicopters in HSI–I regarded as interesting targets in this scene.
2) The Avon Dataset is the second dataset (HSI-II). The Avon dataset was constructed during the data collection campaign, SpecTIR Hyperspectral Airborne Experiment 2012 (SHARE 2012) [48], where a push-broom hyperspectral ProSpec TIR-VS sensor was used. The region of the driving park in Avon, south of Rochester, NY, USA (100 × 100 pixels), was selected for experiments among the morning reflectance data. The dataset includes 360 bands of a spectral range: 400–2450 nm and the spectral resolution: 5 nm. The detection targets include three red or blue felts, 25 tarps with 68 target pixels in a scene.
3) The HYDICE Urban Dataset is the third dataset (HSI-III). The HYDICE airborne sensor is used to collect the data of an urban area, CA, USA. The bands with low SNR and vapor absorption were removed, and consequently, 175 bands were used in the anomaly detection experiment. The image size for each band is 100 × 100. The 21 objects are considered anomaly targets in this dataset.
4) The Fired Mountain Dataset is the fourth dataset (HSI-IV), which was acquired by the AVIRIS sensor with a spatial resolution of 14.7 m. The HSI image was taken in the Arizona fire in the southwestern United States. The burning place in the scene corresponds to 26 pixels of the image as the anomaly targets.

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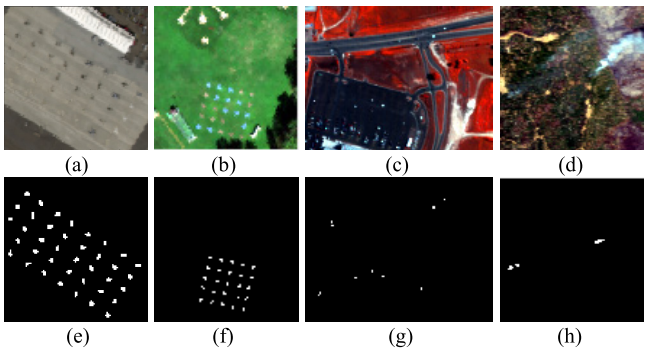


Fig. 6. Examples of HSI datasets I–IV. (a)–(d) False-color images of HSI-I, HSI-II, HSI-III, and HSI-IV. (e)–(h) Corresponding ground truths for (a)–(d).

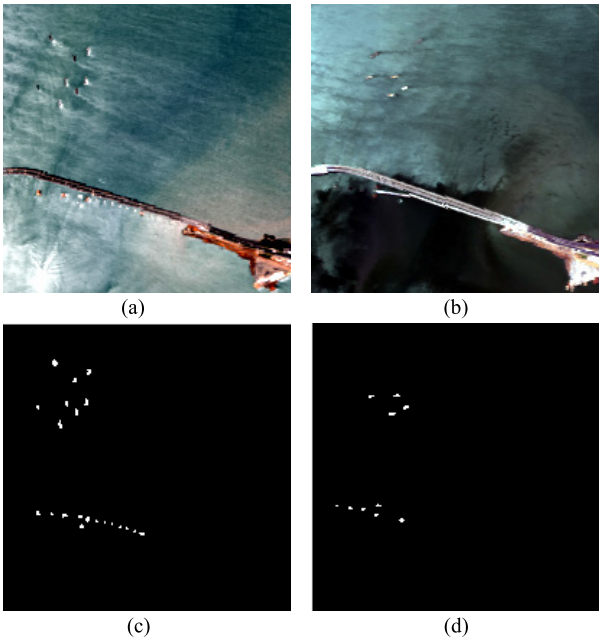


Fig. 7. Examples of HSI datasets V and VI. (a) and (b) False-color images of HSI-V and HSI-VI. (c) and (d) Corresponding ground truths for (a) and (b).

5) The Port Dataset is the last dataset (HSI-V and HSI-VI), which was also captured by the AVIRIS sensor at the San Francisco Bay area port. The dataset contains two hyperspectral images, which are port areas in different time phases. The two images contain ships in different states as anomaly targets. The states of ships are mainly floating on the water surface and diving into the sea. The noisy bands in the original HSIs have been removed.

Examples of the anomaly targets with the corresponding false-color images are depicted in Figs. 6 and 7, and the six datasets are summarized in detail in Table I.

TABLE I
DETAILED INFORMATION OF SIX HSIS

HSI	Size	Capturing location	Spatial resolution	Sensor
HSI-I	100×100	San Diego	3.5m	AVIRIS
HSI -II	100×100	Rochester	1m	ProSpec
HSI -III	100×130	Urban	1m	HYDICE
HSI -IV	100× 100	Arizona	14.7m	AVIRIS
HSI -V	200×200	San Francisco	16.7m	AVIRIS
HSI -VI	200×200	San Francisco	16.2m	AVIRIS

B. Compared Methods and Evaluation Criteria

In the experiment, the proposed method is compared with other hyperspectral anomaly detection methods to verify the effectiveness and robustness of the proposed method. The compared methods are LRX [14], LKRX [16], CRD [29], PAB-DC [25], LRaSMD [24], LCKRX [17], TVSDM [33], and FrFE-RX [34]. The LRX method performs well in HSIs representing simple scenes, which is consistent with the literature. The LKRX and LCKRX expand a linear non-Gaussian model in the low-dimensional original data space to a nonlinear Gaussian domain in higher dimensional feature space. They provide less computational complexity and better accuracy than KRX and LCKRX. The CRD method achieves the collaboration via a linear combination of neighboring pixels of each pixel, and the regularization parameter was set to 10^{-6}, following [29], in this article. For the LRaSMD method, the HSI was decomposed into the anomaly part and the background part, where the background part is represented by a dictionary multiplying the corresponding coefficient matrix and the regularization parameters are set as $\lambda = 0.01$ and $\beta = 0.1$. The PAB-DC and TVSDM also perform well for anomaly target enhancement and noise suppression. Their optimal parameters are used as in [25] and [33].

The detection performances of different detection methods are qualitatively and quantitatively compared in terms of the receiver operating characteristics (ROCs) curve [49] and the area under the ROC curve (AUC) value [50]. A trade-off is inevitable between the probability of detection P_{D} and the false alarm rate P_{FA} during the performance evaluation since a low threshold gives a high P_{D} while a high threshold gives low P_{FA}. The ROC curve is a typical way that fairly describes the trade-off between P_{D} and P_{FA} under the Neyman–Pearson criterion. An increase in P_{D} while maintaining P_{FA} is commonly considered the better performance of the detector. Also, the AUC value is used as a quantitative summary of the ROC curve. When the AUC value of the detection result is larger, the corresponding detection effect is better.

C. Parameter Analysis

In this section, the influence of parameters on anomaly detection performance is discussed, and the optimal parameters are determined by the AUC value. In the process of spatial–spectral constraint detection, three parameters are involved: cluster number k, parameter v, and the window size.

1) Analysis of the Cluster Number k: The number of clusters k of the suspected anomaly target pixels is predefined in the k-means method. The cluster centroid that contains the largest number of detected anomaly target pixels is selected as the representative pixel due to false alarm targets, which is used to guide the optimization process of FrFT order p. AUC values obtained by different clustering numbers k are shown in Table II.

2) Analysis of the Parameter v: The parameter v mainly represents the content of the anomaly target relative to the background feature in HSI. Since there are a limited number

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TABLE II

CLUSTER NUMBER k VALUE OF THE SIX IMAGES

k	3	4	5	6	7	8
HSI-I	0.9432	0.9675	0.9890	0.9768	0.9685	0.9647
HSI-II	0.9876	0.9926	0.9859	0.9762	0.9737	0.9728
HSI-III	0.9637	0.9763	0.9846	0.9904	0.9943	0.9896
HSI-IV	0.9276	0.9527	0.9743	0.9821	0.9866	0.9908
HSI-V	0.9658	0.9726	0.9836	0.9875	0.9827	0.9781
HSI-VI	0.9635	0.9834	0.9942	0.9986	0.9924	0.9815

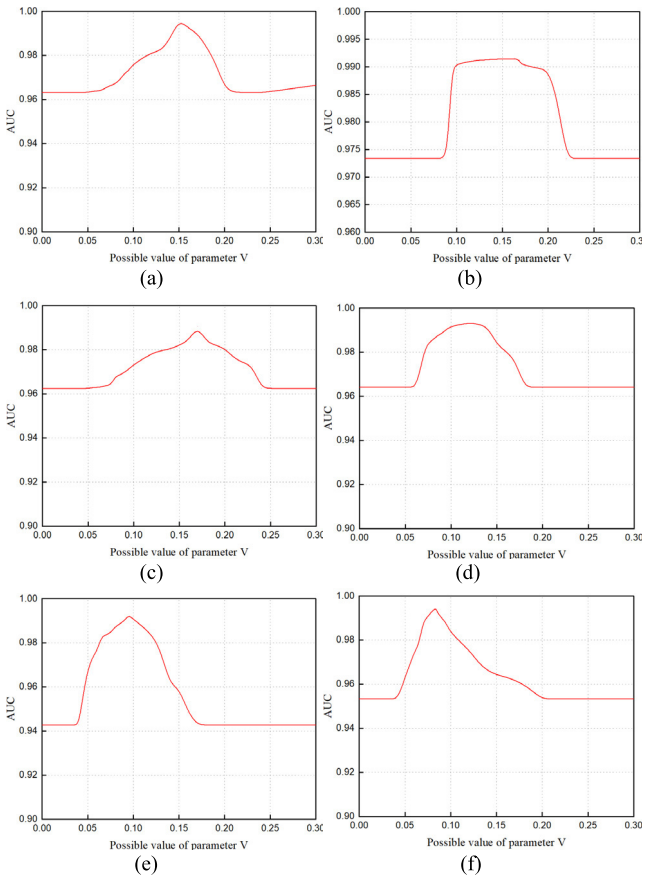


Fig. 8. AUC results according to different values of parameter υ for six datasets. (a) HSI-I. (b) HSI-II. (c) HSI-III. (d) HSI-IV. (e) HSI-V. (f) HSI-VI.

TABLE III

FINAL PARAMETER υ VALUE OF THE SIX IMAGES

υ	HSI-I	HSI-II	HSI-III	HSI-IV	HSI-V	HSI-VI
	0.1527	0.1632	0.1708	0.1235	0.0954	0.0842

of anomaly targets in the image, the distribution of targets is sparse. The AUC values obtained with different parameter υ values are shown in Fig. 8.

Fig. 8 shows that the optimal value of parameter υ that provides the highest AUC is varied for different HSI datasets. Since the content of anomaly targets in each HSI dataset is varied, determining the optimal parameter υ should consider the different characteristics of the HSI and Table III shows the best parameter υ settings for the HSIs.

3) Analysis of the Window Size: Another important parameter is the window size. The proposed method has window parameters with other methods. In LRX, CRD, LKRX, LCKRX, and the proposed method, the window size should be adjusted since it is affected by the anomaly target size. The optimal window size is determined through empirical experiments in this section. The anomaly detection accuracy with different window sizes is compared in Tables IV–IX for the datasets HSI-I to HSI-VI in terms of AUC.

TABLE IV

ANOMALY DETECTION ACCURACY FOR DIFFERENT WINDOW SIZES IN TERMS OF AUC (HSI-I DATASET)

	(3,5)	(5,7)	(7,9)	(7,13)
SASC	0.9567	0.9632	0.9815	**0.9890**
LRX	0.8246	0.8427	0.8653	**0.8870**
CRD	0.9134	0.9384	0.9487	**0.9530**
LKRX	0.8786	0.8932	0.9194	**0.9375**
LCKRX	0.9034	0.9274	0.9365	**0.9428**

TABLE V

ANOMALY DETECTION ACCURACY FOR DIFFERENT WINDOW SIZES IN TERMS OF AUC (HSI-II DATASET)

	(3,5)	(5,7)	(7,9)	(7,13)
SASC	**0.9926**	0.9925	0.9862	0.9766
LRX	**0.9205**	0.9168	0.9081	0.8635
CRD	**0.9605**	0.9562	0.9435	0.9264
LKRX	**0.9537**	0.9465	0.9173	0.8952
LCKRX	**0.9576**	0.9487	0.9256	0.9037

TABLE VI

ANOMALY DETECTION ACCURACY FOR DIFFERENT WINDOW SIZES IN TERMS OF AUC (HSI-III DATASET)

	(3,5)	(5,7)	(7,9)	(7,13)
SASC	0.9819	**0.9943**	0.9184	0.9144
LRX	0.9257	**0.9452**	0.9363	0.9168
CRD	0.9647	**0.9820**	0.9778	0.9576
LKRX	0.9592	**0.9828**	0.9732	0.9275
LCKRX	0.9673	**0.9865**	0.9726	0.9463

TABLE VII

ANOMALY DETECTION ACCURACY FOR DIFFERENT WINDOW SIZES IN TERMS OF AUC (HSI-IV DATASET)

	(3,5)	(7,9)	(7,13)	(15,17)
SASC	0.9107	0.9542	0.9872	**0.9908**
LRX	0.7532	0.8026	0.8432	**0.8991**
CRD	0.8864	0.9029	0.9432	**0.9763**
LKRX	0.8321	0.8576	0.9252	**0.9464**
LCKRX	0.8675	0.8936	0.9257	**0.9572**

TABLE VIII

ANOMALY DETECTION ACCURACY FOR DIFFERENT WINDOW SIZES IN TERMS OF AUC (HSI-V DATASET)

	(3,5)	(5,7)	(7,9)	(7,13)
SASC	0.9864	**0.9875**	0.9732	0.9662
LRX	0.9465	**0.9625**	0.9568	0.9544
CRD	0.9325	**0.9657**	0.9623	0.9537
LKRX	0.9275	**0.9758**	0.9532	0.9484
LCKRX	0.9362	**0.9794**	0.9628	0.9553

TABLE IX

ANOMALY DETECTION ACCURACY FOR DIFFERENT WINDOW SIZES IN TERMS OF AUC (HSI-VI DATASET)

	(3,5)	(5,7)	(7,9)	(7,13)
SASC	0.9873	**0.9986**	0.9924	0.9832
LRX	0.9732	**0.9871**	0.9793	0.9749
CRD	0.9681	**0.9732**	0.9654	0.9561
LKRX	0.9545	**0.9885**	0.9673	0.9264
LCKRX	0.9657	**0.9927**	0.9735	0.9368

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From Tables IV–IX, it can be seen that the detection performance of local anomaly detectors is sensitive to the window size (w_{in} and w_{out}). The optimal window size for each local anomaly detector is determined through the analysis of the effects on the AUC metric with the varying w_{in} (from 3 to 17) and w_{out} (from 5 to 21). The results of several typical window setting experiments are summarized in Tables IV–IX, showing the optimal window setting (bold in Tables IV–IX).

The AUC values of the proposed SASC method are higher than the other methods on different HSI datasets. It indicates that the proposed method can effectively enhance the target while suppressing the background and hence can be useful in detecting and tracking anomaly targets.

D. Anomaly Detection Performance

In this section, the performance of the proposed SASC method is evaluated and compared with eight conventional hyperspectral anomaly detection methods: LRX, LKRX, CRD, PAB-DC, LRaSMD, LCKRX, TVSDM, and FrFE-RX. Fig. 9 qualitatively shows the comparison of the anomaly detection results on six HSI datasets with the optimal parameter settings.

The results show that the proposed method can accurately detect the anomaly targets in different scene images, including airport, city, mountain, and coast. Importantly, the proposed method can detect different sizes of objects, especially small targets and subpixel targets. Due to such advantages, the proposed SASC method can accurately find the anomaly target location, which is useful for battlefield defense, forest-fire prevention, and other practical applications.

Fig. 9 shows that the proposed method can effectively detect spectral anomaly targets compared with the comparison methods. Also, in detecting spectral anomaly targets, the local method provides better performance than the global method.

Besides, the images from HSI-IV to HSI-VI mainly contain strategic targets. When the spatial characteristics are not prominent, the algorithm can enhance the anomaly information through the feedback mechanism introduced by spectral constraint. As a result, the burning point is accurately located in the forest. The ships in different states on the coast are also detected well, such as the ships berthing on the coast and the ships in the water. Those results show the practical value of HSI anomaly detection.

ROC curves of the eight comparison methods on different hyperspectral datasets are depicted in Fig. 10, showing that the SASC method has a lower false alarm rate and higher detection rate over the LRX, LKRX, CRD, PAB-DC, LRaSMD, LCKRX, TVSDM, and FrFE-RX methods. This observation confirms that the SASC method can be applied to anomaly detection in different scenarios. As shown in Table X, compared with other comparison algorithms, the proposed method provides a higher anomaly detection accuracy with a higher AUC value.

Table XI shows the false alarm rates of the compared methods for the fixed detection rate of 0.9. Under the same detection rate, the lower the false alarm rate indicates that the background suppression works the better. The detection rate of 0.9 can be considered that the majority of anomaly targets are detected. Thus, in this article, the detection rate is fixed as 0.9 when the false alarm rate is compared for each method. It can be seen that the proposed method can significantly suppress the background and reduce the false alarm in the results, ensuring the detection of anomaly targets in HSI.

In the FrFE-RX, the FrFT is used as a transformation itself, where the traditional RX method is directly applied to the transformed images for anomaly detection. FrFE-RX obtains the background and anomaly parts in the hyperspectral image (I + II + III) (Fig. 4) through the FrFT. The divided parts are used to suppress the noise, thereby improving the performance of the RX algorithm. However, it does not fully suppress the background, nor does it take into account the spectral anomaly targets contained in the image. Unlike the FrFE-RX method, the proposed method utilizes the FrFT to link the spatial constraint to the spectral constraint. The information represented by the spatial anomaly results is used as the guide information to determine the optimal order of the FrFT. With the FrFT, the suspected anomaly part (II + III) (Fig. 4) is obtained, which not only suppresses noise also fully suppresses the background and highlights the target. It is conducive to the subsequent spectral constraints and feedback process. It is oriented to the spectral anomaly targets. The experimental results show that the proposed method can be applied to the spectral anomaly targets as well as the spatial targets. The detection performance of anomalous targets also is improved, proving the innovation and advantages of the proposed method.

Table XII lists the comparison of the computation time (s) of the compared methods. All experiments were carried on a machine with an Intel Core i7-7700 CPU and an 8-GB RAM.

Global methods usually provide less computation time than local ones. The PAB-DC method uses the whole scene to build a background dictionary and potential exception dictionary, incurs an enormous computational burden, and consumes more time.

E. Influence Analysis of Spectral Constraint

In order to verify the contribution of spectral constraint in anomaly detection, the AUC value of anomaly target candidates obtained by spatial constraint is compared with that of anomaly detection results obtained by the proposed method.

Table XIII shows that the feedback mechanism introduced by spectral constraint can effectively improve the detection accuracy while reducing the false alarm rate. In order to make the suspected anomaly target part include all the anomaly targets in the image, increasing false alarms in the anomaly target candidates, the spectral constraint considers the spectral changes of the anomaly target candidates and the surrounding background.

Such spectral changes are caused by spectral anomalies, as well as spatial anomalies, which are so-called camouflage targets, hidden targets, and deceptive targets [51]. When the anomaly characteristics are not apparent in the spatial domain, but the environmental change causes subtle spectral changes, spectral anomalies are induced, such as the burning point. The smoke coverage area has anomaly characteristics in space,

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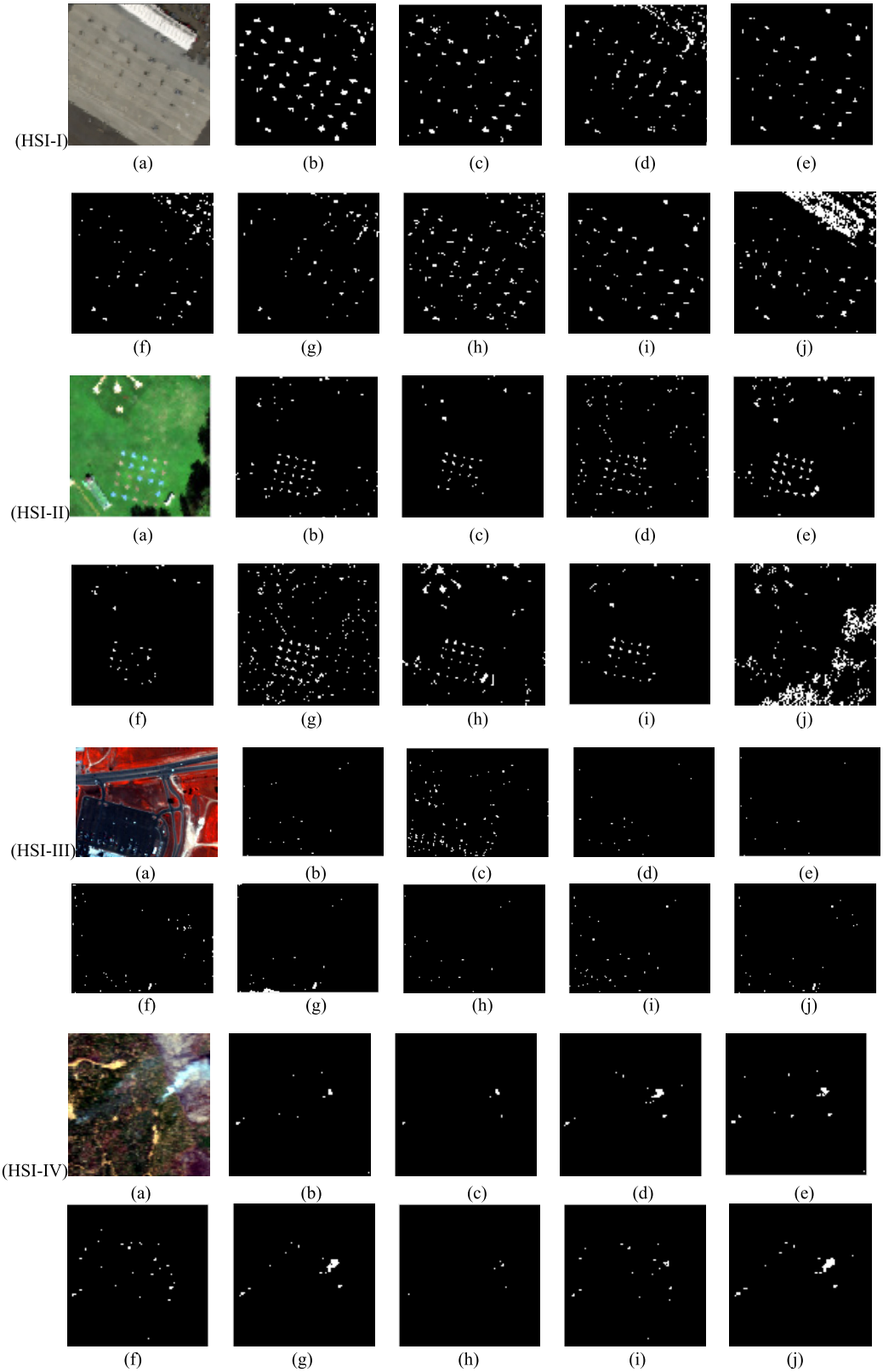


Fig. 9. Anomaly detection performance results for the compared methods on six HSI datasets. (a) Input image. (b) Proposed method. (c) LRX. (d) LKRX. (e) CRD. (f) PAB-DC. (g) LRaSMD. (h) LCKRX. (i) TVSDM. (j) FrFE-RX.

but the temperature difference leads to spectral differences. From the perspective of the application, accurate detection of the combustion point can be conducive to mountain fire protection, bring substantial application value. HSI-V and HSI-VI contain the ship diving into the water, where the spectrum of the area is partially changed due to the entry

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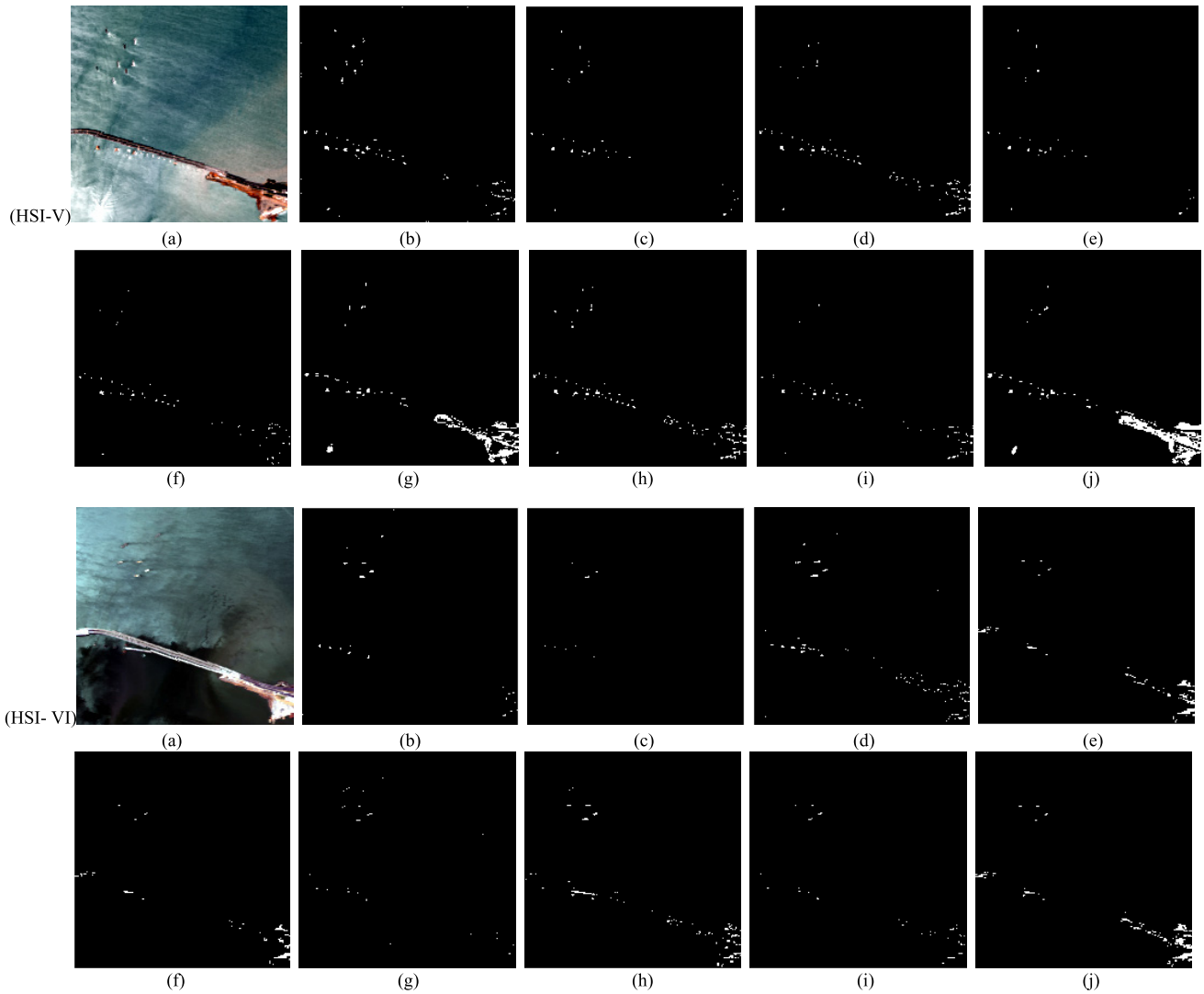


Fig. 9. *(Continued.)* Anomaly detection performance results for the compared methods on six HSI datasets. (a) Input image. (b) Proposed method. (c) LRX. (d) LKRX. (e) CRD. (f) PAB-DC. (g) LRaSMD. (h) LCKRX. (i) TVSDM. (j) FrFE-RX.

TABLE X

AUC COMPARISON OF DIFFERENT METHODS FOR ANOMALY DETECTION PERFORMANCE ON SIX DATASETS

HSIs	SASC	LRX	LKRX	CRD	PAB-DC	LRaSMD	LCKRX	TVSDM	FrFE-RX
HSI-I	**0.9890**	0.8870	0.9375	0.9530	0.6407	0.6363	0.9428	0.9643	0.7984
HSI-II	**0.9926**	0.9205	0.9537	0.9605	0.8975	0.9742	0.9576	0.9761	0.6237
HSI-III	**0.9943**	0.9357	0.9828	0.9820	0.9362	0.9809	0.9865	0.9847	0.9800
HSI-IV	**0.9908**	0.8991	0.9464	0.9763	0.8413	0.9521	0.9572	0.9578	0.9579
HSI-V	**0.9875**	0.9625	0.9758	0.9657	0.9239	0.8934	0.9794	0.9326	0.9454
HSI-VI	**0.9986**	0.9871	0.9885	0.9732	0.8404	0.9538	0.9927	0.9475	0.9735

TABLE XI

FALSE-ALARM RATES COMPARISON OF DIFFERENT METHODS ON SIX DATASETS

HSIs	SASC	LRX	LKRX	CRD	PAB-DC	LRaSMD	LCKRX	TVSDM	FrFE-RX
HSI-I	**0.035**	0.2016	0.085	0.06	0.7625	0.75	0.065	0.045	0.5
HSI-II	**0.02**	0.0705	0.06	0.047	0.2567	0.043	0.055	0.053	0.87
HSI-III	**0.0217**	0.3025	0.042	0.035	0.06893	0.036	0.037	0.04	0.048
HSI-IV	**0.0345**	0.1967	0.065	0.0665	0.2834	0.0679	0.054	0.053	0.093
HSI-V	**0.045**	0.075	0.064	0.055	0.08594	0.1467	0.062	0.067	0.095
HSI-VI	**0.0452**	0.0504	0.05	0.065	0.08762	0.0586	0.045	0.07	0.062

of the submarine. The anomaly target candidates detected by the spatial constraint potentially include many false alarms. The feedback mechanism based on the spectral constraint can detect and track the spectral anomaly target.

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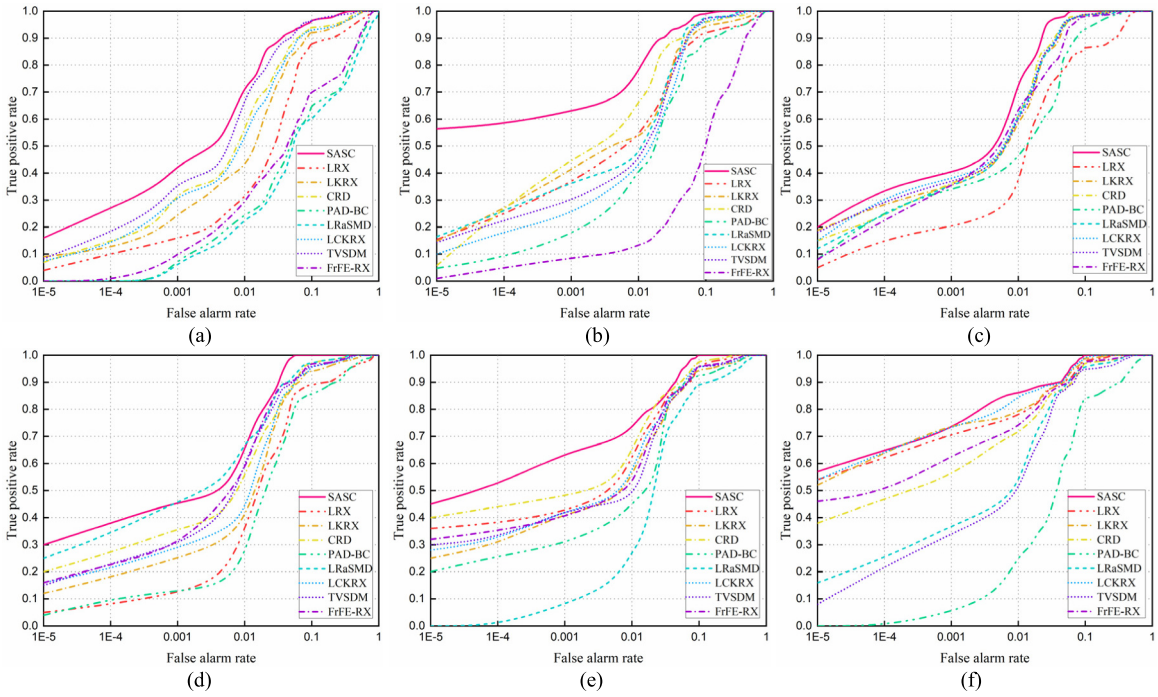


Fig. 10. ROC curves comparisons for different methods on (a) HSI-I, (b) HSI-II, (c) HSI-III, (d) HSI-IV, (e) HSI-V, and (f) HSI-VI.

TABLE XII
COMPUTATION TIMES ON SIX DATASETS WITH DIFFERENT METHODS AND THE OPTIMAL WINDOW SIZE

HSIs	SASC	LRX	LKRX	CRD	PAB-DC	LRaSMD	LCKRX	TVSDM	FrFE-RX
HSI-I	75.37	39.42	201.37	67.48	93.21	35.83	186.75	60.27	10.91
HSI-II	74.21	28.45	89.65	37.65	95.37	26.76	76.86	57.63	72.99
HSI-III	86.52	45.27	37.42	94.56	99.53	42.34	13.48	75.86	17.47
HSI-IV	72.55	34.38	45.38	62.42	92.25	38.32	25.76	83.57	19.79
HSI-V	267.46	126.45	586.96	225.18	367.24	124.17	525.75	156.34	72.95
HSI-VI	265.34	124.57	637.57	243.26	367.38	135.52	568.43	167.86	74.98

TABLE XIII
INFLUENCE OF THE FEEDBACK

	HSI-I	HSI-II	HSI-III	HSI-IV	HSI-V	HSI-VI
Proposed method	0.9890	0.9926	0.9943	0.9908	0.9875	0.9986
Without feedback	0.9423	0.9362	0.9467	0.8865	0.9217	0.9576

TABLE XIV
INFLUENCE OF THE FrFT

	HSI-I	HSI-II	HSI-III	HSI-IV	HSI-V	HSI-VI
Proposed method	0.9890	0.9926	0.9943	0.9908	0.9875	0.9986
Without FrFT	0.9825	0.9842	0.9836	0.9543	0.9672	0.9585

A similar conclusion is drawn from the ROC curve in Fig. 11. It is precise due to the feedback network constrained by spectral information that the false alarms contained in the suspected anomaly targets can be removed, and the anomaly targets contained in the image can be accurately found.

F. Influence Analysis of Fractional Fourier Transform

In this section, the difference-amplification of FrFT on target and background is verified by comparing AUC values.

Table XIV lists the comparison of the AUC values of the detection results of this method in the above two different cases with and without FrFT. As shown in Table XII, the FrFT effectively amplifies the difference between the target and the background, especially in the detection of spectral anomaly targets. From the comparison of HSI-IV to HSI-V, it is found that the use of FrFT has a significant impact on the spectral anomaly target detection performance. FrFT can enlarge these differences, conducive to the spectral constraint process and feedback process. After comparing HSI-I to HSI-III, it is found that the use of FrFT has little effect on the spatial anomaly target detection performance. SASCs are enough to improve the anomaly detection performance. The amplification of the difference between the target and the background by the FrFT does not significantly contribute to the spectral constraint process. Therefore, the use of FrFT in the spectral anomaly target detection has a more obvious performance improvement, while the detection of spatial anomaly targets can improve the detection performance, but the improvement effect is not apparent.

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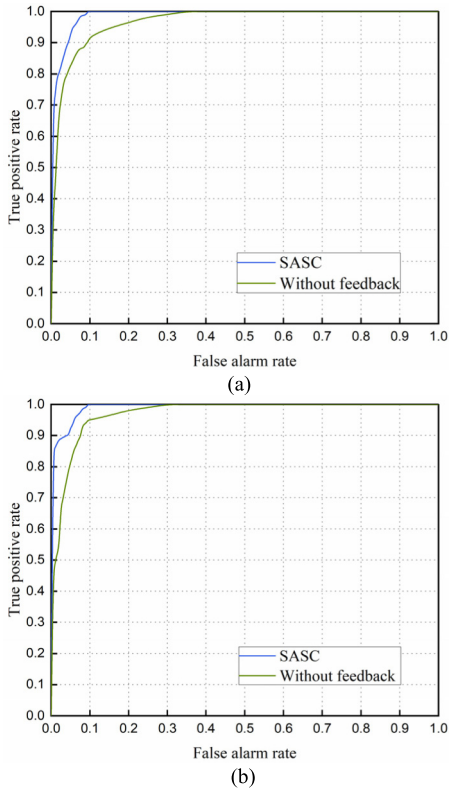


Fig. 11. ROC curves of the SASC method with and without feedback on the port datasets. (a) HSI-V. (b) HSI-VI.

IV. Conclusion

This article proposes a novel hyperspectral anomaly detection method for spectral anomaly targets based on SASCs. Furthermore, a new strategy is introduced to determine the optimal order of FrFT by combining the spatial anomaly and the generalized uncertain principle. Experimental results on real HSI datasets show that the proposed SASC method can effectively suppress the background and enhance the detection accuracy. Moreover, the proposed method can accurately detect the spectral anomaly targets as well as spatial anomaly targets. It highlights the practical applicability of the proposed method.

One relevant topic deserving further research is selecting the region of interest (ROI) for hyperspectral anomaly detection. Most of the detection methods including the proposed method are designed for HSI of small-scale. Thus, the project of improving the proposed method on how to select the ROI, containing anomaly targets from large-scale HSI, is interesting as well as challenging in the future work.

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